

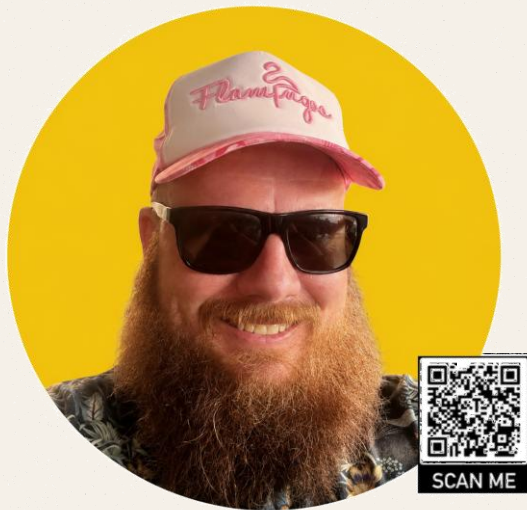


DATA POINT

Prague

Dataflows, once known as M or Power Query, still reign supreme in Citizen Data Engineering.

Empower people to truly enjoy data with M and Power Query — the tools that make data easy, intuitive, and powerful. Learn how M-based transformations help professional developers support and elevate citizen developers beyond Excel, build consistent and governed models, and create a strong data culture where transformation drives collaboration and success.



Michael Tenner

The Data Enthusiast · Full Stack Power BI Engineer

ABOUT HIM

Poker hobbyist and craft beer enthusiast. He lives in Cologne when he's not trying to get to Easter Island or wondering whether you can really eat that.

Daten-WG · Knowledge Kitchen · Cologne

DATAPOINT PRAGUE 2026 · SESSION BY MICHAEL TENNER

Source · Transform · Model

Why M and Power Query **still reign supreme**
in Citizen Data Engineering

Michael Tenner

Daten-WG · **Knowledge Kitchen** · Full Stack Power BI Engineer

20 minutes · English · Live demo

Thank you for making this possible



Every Power BI consultant has seen **this ticket.**

And every fix attempted in fifteen years of governance has missed the point.

Report_FINAL_v7_REAL_THIS_ONE.xlsx

Sales by Region (Q1–Q4)				
	Region	Amount	Amount_new	Q

- 01 Merged cells, headers in row 4.
issue
- 02 Two columns: Amount and Amount_new.
issue
- 03 Four sheets named “do not touch.”
issue

Shadow BI grows where IT says no. This talk is about why **M** is the better answer than policy.

	A	B	C	D	E	F	G	H	I	J	K	L	N	O	P	Q
1	SALES ANALYSIS															
2	in kEUR															
3	REGION	PRODUCT GROUP	ACTUALS			BUDGET			VARIANCE							
5			Q1	Q2	H1	Q1	Q2	H1	H1							
									abs.	%						
6	Europe	Product A	1200	1350	2550	1100	1300	2400	150	6,3%						
7		Product B	900	950	1.850	850	900	1.750	100	5,7%						
8		Product C	600	700	1.300	650	750	1.400	-100	-7,1%						
10		Europe Total	2.700	3.000	5.700	2.600	2.950	5.550	150	2,7%						
11	Americas	Product A	800	850	1.650	750	900	1.650	0	0,0%						
12		Product B	700	750	1.450	600	700	1.300	150	11,5%						
13		Product C	400	500	900	450	500	950	-50	-5,3%						
14		Americas Total	1.900	2.100	4.000	1.800	2.100	3.900	100	2,6%						
15	Grand Total		4.600	5.100	9.700	4.400	5.050	9.450	250	2,6%						

BI DEVELOPER

MERGED CELLS EVERYWHERE!

HARD-CODED FORMULAS...

FORMATTING FROM HELL!

DATA NEVER SLEEPS

I CREATE STRUCTURE.
I PREVENT CHAOS.

IF ONE ROW CHANGES, EVERYTHING BREAKS.

WHO NAMED THIS RANGE?

FINAL_v3_REAL_FINAL.xls

MANUAL UPDATES.
MANUAL RISKS.

CONTROLLER

PERFECT OVERVIEW ON ONE PAGE!

EXACTLY WHAT I NEED FOR MY REPORTS.

EASY TO READ AND UNDERSTAND.

I CAN SEND IT TO ANYONE.

I KNOW EXCEL. I TRUST IT.

I NEED CLARITY.
I MAKE DECISIONS.

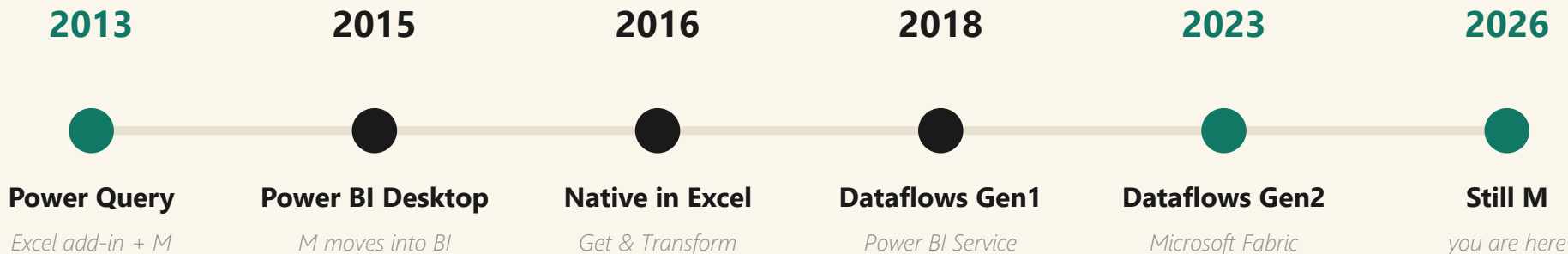


Same table. Different pain.
Same goal: better numbers, better decisions.

Every BI professional has seen **dreamed about this...**

M has been the constant for **thirteen years** — outlived every UI it lived in.

What is M? *The functional formula language Power Query speaks.*



M ALSO LIVES IN

Power Apps · Power Automate · **Azure Data Factory** ·
Fabric Data Factory · SSIS · SSAS · Customer Insights ·
Dataverse

THE HIDDEN SUPERPOWER · QUERY FOLDING

M translates to native SQL and pushes work **down to the source.**

Fully folded queries run very fast
(source: Microsoft Learn)

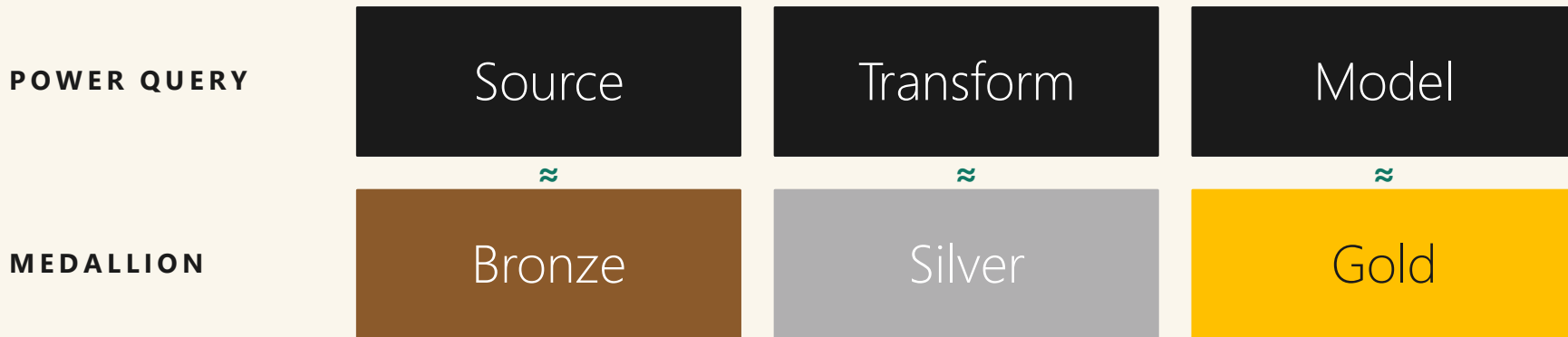
Visual ETL tools have existed **since the 1990s**. Most ended up where they didn't belong.

Tool	Since	Built for	What actually happened
Informatica	1993	Enterprise ETL	Locked into IT. Business stayed out by design.
Alteryx	1997	Analyst data prep	Loved by analysts. Per-seat pricing kept it niche.
KNIME	2004	Data science workflows	Often misused as ETL — it's a science tool.
SSIS	2005	SQL Server pipelines	Misused for business logic that belonged in stored procs.
Talend	2005	Open-source big-data ETL	Citizen-friendly in theory. IT-only in practice.
Tableau Prep	2018	Tableau users' prep	Useful, but never escaped the Tableau ecosystem.
Power Query	2013	Excel users	First tool that actually empowered citizen developers.

They all promised *"self-service."* Only one delivered — because it started **inside Excel**.

Power Query already speaks **Medallion** — just point it out.

What is Medallion? *A three-layer pattern that separates raw, refined, and ready-to-consume data.*



Core Warehouse → Medallion → Power Query. Same instinct: separate raw, refined, and ready.

Three properties make Power Query **different** — and they are exactly what AI cannot fake.

01

Named

Every transformation has a label business users can read — no training required.

AI: outputs code, no narrative

02

Ordered

Steps run in sequence. Reorder, insert, remove — see what changes immediately.

AI: regenerates whole block

03

Replayable

Refresh re-runs every step. The logic IS the documentation.

AI: regenerates, drift creeps in

LIVE DEMO · 10 MINUTES

From an ugly Excel crosstab to a governed Dataflow.

Tell — Show — Tell, in three cycles

A

Source / Stage / Bronze

Connect, don't clean

2 min

B

Transform / Core / Silver

Unpivot, type, enrich

5 min

C

Model / Presentation / Gold

Flat → Star, Dataflow

3 min

The raw layer is **sacred** — connect to it, but never fix anything there.

If we surgery the source, we lose lineage forever.

Same layer, three names: Source · Stage · Bronze

01

Connect, don't clean

Raw mess in, as-is.
No type fixes here.

02

Make origin obvious

File path, sheet, query.
Lineage starts at Bronze.

03

Leave the mess visible

Show what Power Query
actually sees on import.

ON SCREEN: Open the .xlsx in Power Query · point at Applied Steps · no transformations yet

Each named step is a **teaching moment** — dimensional thinking enters quietly.

Business users learn the vocabulary by watching what each step is called.

Same layer, three names: Transform · Core · Silver



Each step appears in the Applied Steps pane, named in plain English. The business user reads:

“Ah — first headers, then we pivot, then we type, then we calculate.”

- **Auditable**
Every step is named.
- **Editable**
Reorder, insert, delete.
- **Self-documenting**
The logic IS the documentation.

ON SCREEN: Click each step · show Advanced Editor briefly · “you don’t have to write M, but you can”

Save as **Dataflow** — one source feeds Power BI, Excel, and beyond.

Same layer, three names: Model · Presentation · Gold

STEP 1

Flat Table

Not perfect. Immediately usable.



STEP 2

Star Schema

Split out dimensions. Scale.



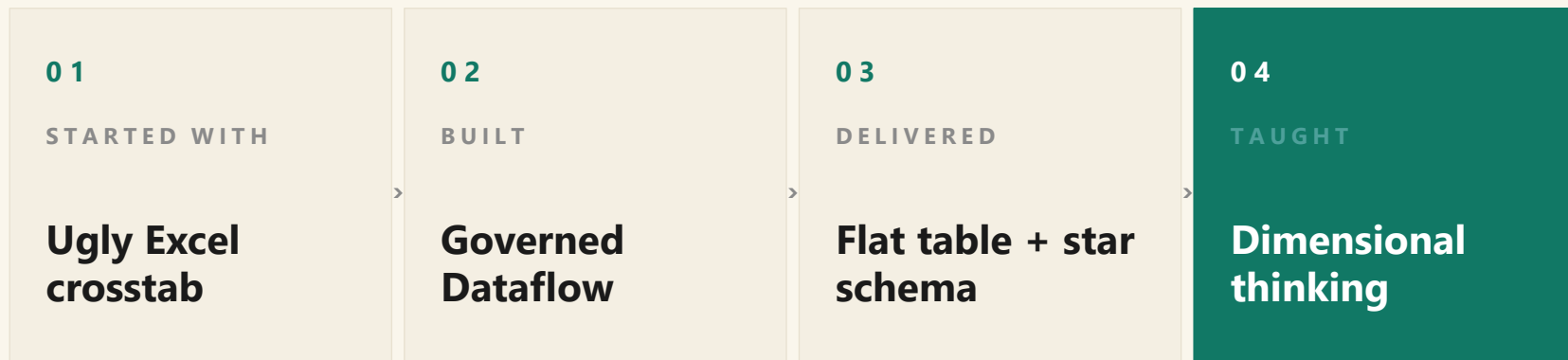
ONE DATAFLOW · MANY CONSUMERS

Power BI Report

Excel Pivot

SQL / Python next

In ten minutes, we taught **dimensional thinking** — without ever using the term.



The Dataflow is never wasted work.

It is either the final product — or the spec for the final product.

Same flat shape. **Different runtime.**

A well-structured Dataflow is the spec for what comes next.

TODAY

Dataflow Gen2

fact_sales (flat)

Named, ordered M steps. Auditable, replayable.



TOMORROW — WHEN THE DATAFLOW EARNS IT

SQL View / Stored Procedure

Database does the work. Push down via Query Folding.

Python Notebook (Fabric)

Pandas, PySpark, ML. Same flat table.

Spark SQL (Lakehouse)

Lakehouse-native. Scale to TB+.

Start flat. Never crosstab in the database.

Crosstabs lock you into one UI shape. Flat tables port to anything.

Three principles to **elevate citizen developers** — without losing governance.

01

Coach — don't gate-keep.

Explain the why. Then point at the Applied Steps pane. Stop writing policies.

02

Promote what works — even imperfectly.

A 70 %-correct Dataflow used by 50 people beats a perfect spec read by none.

03

Graduate at ~10 min refresh or 50+ consumers.

Convert to SQL view or Python notebook. AI helps. The Dataflow was the spec.

AI doesn't replace Power Query — it **amplifies** the people who already speak its language.

"But in 2026, why teach this? I'll just ask an LLM for the Python or SQL."

ANSWER 01

Visibility beats generation.

LLMs produce output. They do not show the business user why step 3 has to happen before step 4. Trust is built by traceable thinking — a UX problem AI hasn't solved.

ANSWER 02

AI needs a vocabulary.

You can't ask AI a question you cannot formulate. Power Query teaches the primitives: pivot, unpivot, merge, group. Then AI is a force multiplier — not a substitute.

AI doesn't replace the ladder. Power Query is still the first rung.

ONE LINE TO REMEMBER

Stop writing policies.
Start showing steps.

This is how data culture forms — not how it's mandated.

Long live M. Long live the citizen data engineer.

Questions?

Michael Tenner · Daten-WG · daten-wg.com · datenwgknowledgekitchen.com

Feedback | Resources & Links

FEEDBACK



Dataflows, once known as M or Power Query, still reign supreme in Citizen Data Engineering.

Scan to share feedback

GITHUB REPO

Data Point Prague Demo

Git repo with file for the
demo

<https://github.com/Losveratos/Data-Point-Prague-Demo>

LINKEDIN

Michael Tenner

Connect with me

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KNOWLEDGE BASE

Knowledge Kitchen

Guide + YouTube

<https://datenwgknowledgekit.chen.com/index.html>

Thanks — let's stay in touch.